

RISC-V Ottawa

Hardware Root of Trust

The final turtle...

Member project pitch · June 17, 2026 · Yusef Karim



What is a root of trust?

The small, trustworthy core the rest of a system's security is built on. It owns **secure boot**, **identity keys**, and **attestation**, and everything else trusts it implicitly.

Security is turtles all the way down.

The root of trust is the final turtle.

It is foundational to every secure device, from a phone to a car to a data-centre server.

Yet almost every one is a black box...you can never knows what's in it or what code it runs.

Now it's open, ALL the way down

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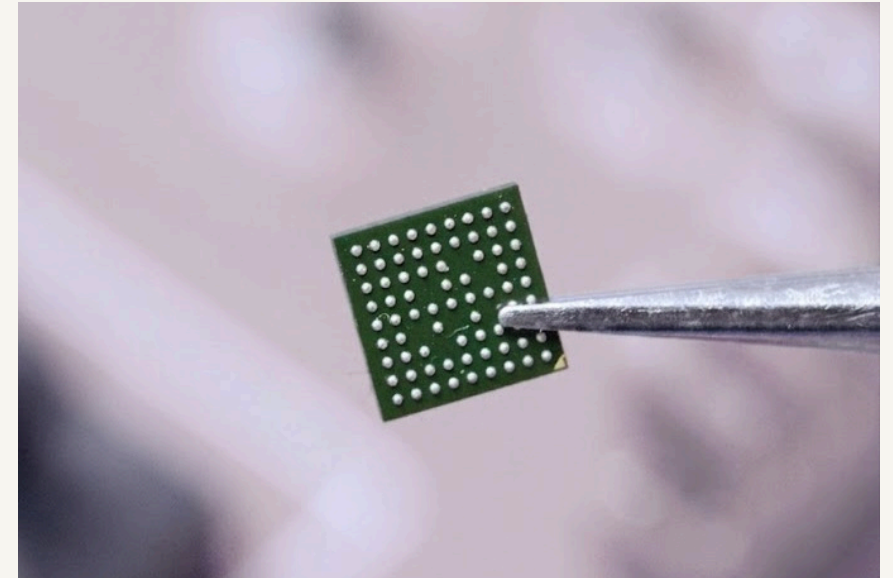
OpenTitan is the first root of trust open **all the way down** – to the actual chip design, on **RISC-V**. As of this year it ships in production **Google Chromebooks**.

OpenTitan

Secure boot, key management, attestation, on **RISC-V**. Runs in **Verilator** simulation, no hardware required to start.

Pavona

The newer, certification-aligned distribution, adds a **post-quantum** crypto stack and a modern key manager.



OpenTitan chip shipping in Chromebooks

Real security on open RISC-V hardware that we can poke and prod.

Why?

I learned about OpenTitan five years ago during my Master's.

Then I finished, became a boring software engineer, and never touched it again.

I've wanted to come back to it ever since. This group is the excuse.

The tooling is free, the implementation is fully open, and there are **so many things to learn** about.

...so many things to learn

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secure boot ML-KEM OTP fuses **attestation** ePMP Ibex **lifecycle** ASCON DICE
side-channel analysis KMAC ownership transfer **Verilator** root keys fault injection **ML-DSA**
flash scrambling **certificate chains** **entropy source** FIPS 140-3 key derivation **SPHINCS+** FPGA bring-up?
measured boot **CSRNG** HMAC tamper detection **ChipWhisperer** signature verification
SRAM scrambling **Bazel** post-quantum crypto EDN **fusesoc** AES threat modeling **DRBG** secure storage
ECIES **provisioning** hardened counters RSA / ECC **Tock OS** embedded Rust CDC / RDC **mailboxes** DMA
Sky130 **TinyTapeout** tapeout AST **redundancy** comfortable IP key manager **DV / UVM** firmware update
rollback protection **TPM vs HSM vs SE** memory protection PKI **TileLink**

The path: make it, then break it

understand it → run it → break it

Simulation-first

Build the chip in **Verilator** on a laptop, no FPGA needed.

Hardware later, if we can get it

The supported FPGA boards (CW310/CW340) are **very** expensive (\$8k–16k).

Understand

Secure boot, identity, attestation, key management.
Subsystem by subsystem.

Run

Run it in **Verilator** and test end to end: provision, boot, attest, run an owner app.

Break

Forge things, poke it in weird ways. Figure out what it can and can't do.

Side quest: bribe someone to get an FPGA

Interested?

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This ties together many things the group already cares about: **RISC-V, simulation, FPGAs, embedded Rust and C, and secure firmware.** Hard to do alone, very doable together.

Now Join **Discord #hwrot**

Learn more Browse opentitan.org and pavona.org.

Help ...if you know anyone who'd lend us a \$16k FPGA, let me know 🙏.

